

An On-Chip Streaming Fixed-Point Vision Processing Pipeline for Epiretinal Prostheses

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Abstract—Epiretinal prostheses restore partial vision by electrically stimulating retinal ganglion cells. Current vision processing units (VPUs) are belt-worn, bulky, and power-hungry, motivating integration within spectacle frames under strict area and power constraints. This work presents a streaming, fixed-point classical computer vision pipeline for a spectacle-mounted VPU targeting epiretinal prostheses. The proposed pipeline converts a 640×480 sRGB camera frame into an enhanced 80×60 grayscale output through seven cascaded stages: colorspace linearization, area-average downsampling, spatial denoising, temporal smoothing, histogram-based contrast enhancement, gamma correction, and feature fusion of edge, saliency, and contrast cues. The architecture restricts frame-level storage to the 80×60 byte state buffer required for temporal smoothing and uses only local line buffers elsewhere, requiring 8.56 kB of total on-chip buffering. The pipeline uses AXI-Stream interfaces and is implemented in TSMC 65 nm CMOS technology, achieving 0.81 mm^2 area and 9.5 mW power at 12.5 MHz. Phosphene-domain evaluation on 452 frames shows higher edge-localized percept brightness and significantly lower inter-frame flicker relative to standard preprocessing baselines.

Index Terms—epiretinal prosthesis, streaming architecture, vision processing unit, fixed-point arithmetic, phosphene simulation

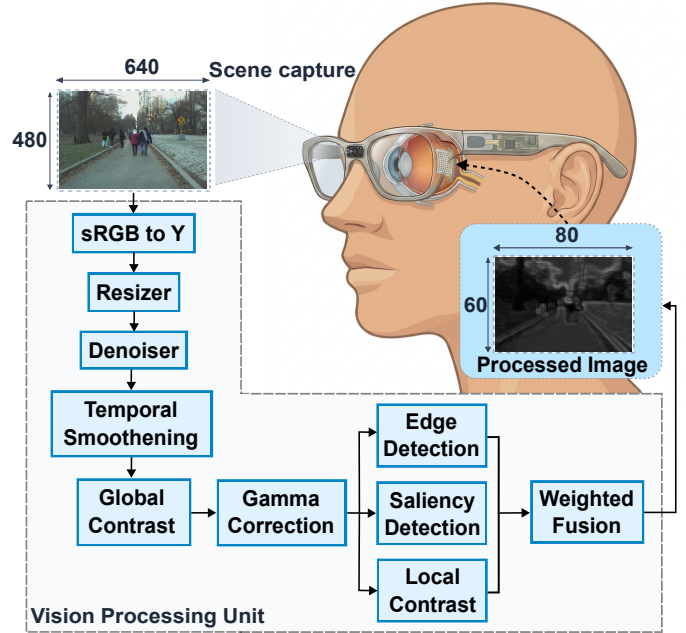


Fig. 1. Proposed VPU pipeline. A 640×480 sRGB frame is converted to an enhanced 80×60 output through fully streaming stages.

I. INTRODUCTION

Retinal degenerative diseases—retinitis pigmentosa (RP) and age-related macular degeneration (AMD)—progressively destroy photoreceptor cells while leaving the inner retinal circuitry (bipolar and ganglion cells) largely intact. AMD alone affects an estimated 196 million people worldwide, projected to reach 288 million by 2040 [1], while RP affects approximately 1 million [2]. Epiretinal prostheses offer a promising intervention by stimulating residual ganglion cells with spatially encoded electrical pulses, generating phosphene percepts that restore coarse functional vision.

Current retinal prosthesis systems, such as Argus II [3] and IRIS II [4], include a belt-worn Vision Processing Unit (VPU) that converts frames into brightness maps for stimulating an implanted electrode array. These external units require regular recharging [5] and their bulk causes persistent user discomfort—both significant barriers to long-term adoption. Next-generation prostheses demand a VPU compact enough to integrate within spectacle frames while operating within a strict power budget, imposing extreme constraints on the underlying image processing hardware.

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Preprocessing visual scenes for retinal prostheses has been an active research direction for over a decade. Early work established that raw camera images require prosthesis-specific transformations, including downsampling, contrast enhancement, and edge emphasis, to generate perceptually meaningful phosphene patterns from low-density electrode arrays [6], [7]. Saliency-driven approaches have since shown that selectively enhancing foreground objects and suppressing background clutter improves the object recognition under simulated prosthetic vision [8], [9]. More recently, edge-focused preprocessing has been demonstrated to improve visual acuity outcomes while simultaneously reducing computational load [10]. Collectively, these works motivate a pipeline that integrates spatial denoising, temporal stabilization, contrast enhancement, and edge-saliency fusion as sequential, purpose-built stages—precisely the architecture proposed here.

Despite this body of algorithmic work, on-chip realizations of complete prosthetic VPU pipelines remain scarce in the open literature. Existing implementations are predominantly FPGA-based, targeting the scene downscaling and electrode mapping [11], or coupling FPGA processing with psychophysical evaluation without systematically addressing power or area constraints [12].

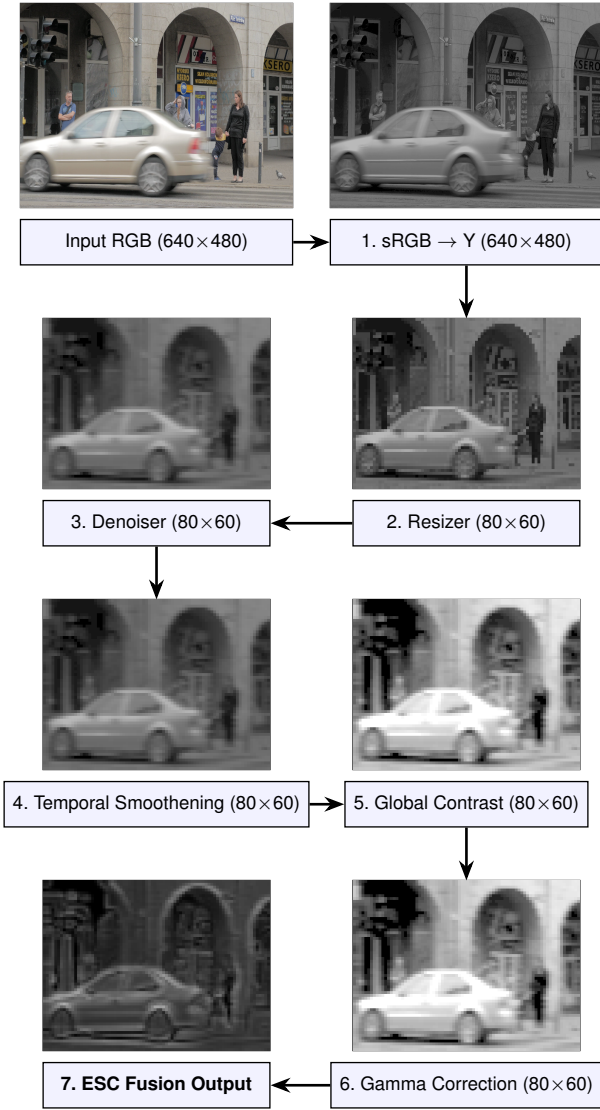


Fig. 2. Proposed hardware pipeline. Display sizes reflect internal resolution transitions (640 × 480 to 80 × 60). The ESC Fusion Output represents a weighted fusion of edge detection, saliency detection, and local contrast features.

This paper presents a classical VPU pipeline for retinal prosthesis preprocessing, with three contributions:

- 1) A seven-stage fully streaming fixed-point pipeline architecture that confines frame-level storage to a single 4.8 kB temporal smoothing buffer and uses only local line buffers elsewhere.
- 2) Hardware-aware on-chip implementation in TSMC 65nm technology, demonstrating feasibility within wearable power and area budgets.
- 3) A quantitative evaluation framework combining image quality metrics and phosphene perception modelling to validate on-chip pipeline output.

II. VPU PIPELINE ARCHITECTURE

A. System Overview

The proposed VPU converts a 640×480 sRGB stream into an enhanced 80×60 grayscale output for electrode stimulation

(Fig. 1). All stages operate in a streaming manner using AXI-Stream interfaces, with pixels processed in raster as shown in Fig. 3. Each module maintains local state, comprising of 3-row line buffers for the spatial filtering stages and a single 4.8 kB frame buffer for the temporal smoothing stage, with no additional full frame buffering in the datapath. Fig. 2 illustrates the output at each stage of the pipeline for a representative outdoor frame.

B. Stage Descriptions

Stage 1 - sRGB to Linear Luminance: RGB888 inputs are first linearized using three 256-entry LUTs, producing linear intensities in Q1.7 format (8-bit fixed-point), and then converted to luminance via fixed-point BT.601: $Y = (27R_{lin} + 91G_{lin} + 9B_{lin} + 64) \gg 7$. This produces a perceptually meaningful intensity signal for downstream processing while reducing computation to a single channel (LUTs require 768B of memory).

Stage 2 - Spatial Downsampling: This stage uses separable $[1 \ 2 \ 1]/4$ Gaussian filter followed by $8 \times$ subsampling to reduce the resolution from 640×480 to 80×60. It is implemented in a streaming fashion with a 1920 B (3×640) line buffer for storing the input and the intermediate pixel values. The 2 MAC units for horizontal and vertical convolution are fed in the appropriate inputs from the row buffer with appropriate padding done for the edges of the image. The filter suppresses aliasing artifacts that would otherwise corrupt low-resolution structure in downstream stages.

Stage 3 - Spatial Denoising: A second $[1 \ 2 \ 1]/4$ Gaussian filter at 80×60 smooths residual downsampling noise, using a 2×80 B line buffer.

Stage 4 - Temporal Smoothing: An adaptive exponential moving average (EMA) filters inter-frame flicker. The blending coefficient α is chosen based on the per-pixel difference between frames: slow α for small changes (stability) and fast α for large changes (responsiveness), with linear interpolation between thresholds:

$$\text{out} = (\alpha \cdot \text{current} + (256 - \alpha) \cdot \text{previous}) \gg 8 \quad (1)$$

A frame buffer stores the previous EMA output for the 80×60 frame.

Stage 5 - Global Histogram Contrast Enhancement: A percentile-based contrast stretch expands pixel values to the full electrode dynamic range. A 256-bin histogram is accumulated for each frame while simultaneously stretching pixels using the previous frame's percentile bounds p_5 and p_{95} , which are also EMA-smoothed by using an IIR to prevent abrupt shifts:

$$\text{out} = \text{clamp} \left(\frac{(\text{in} - p_5) \cdot 255}{p_{95} - p_5}, 0, 255 \right) \quad (2)$$

Division is implemented as a 16-step restoring divider computing a Q8.8 reciprocal. The histogram requires 416 B of memory.

Stage 6 - Gamma Correction: A 256-entry LUT applies a $\gamma = 2.2$ expansion curve to restore perceptual luminance linearity for downstream processing, requiring 256 B of memory.

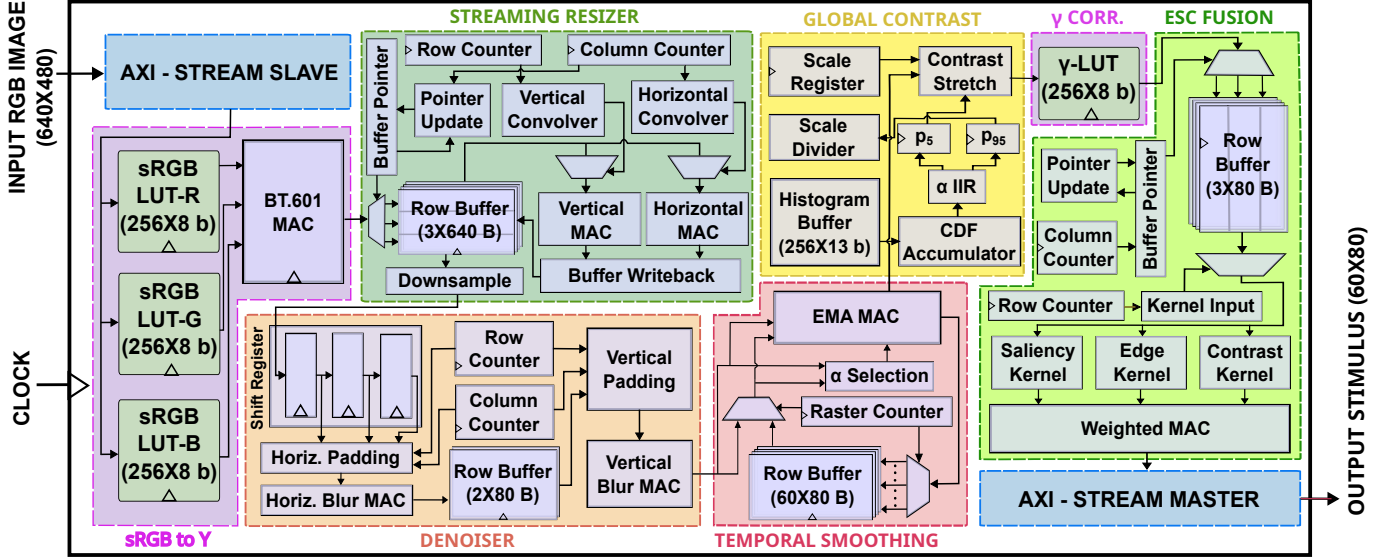


Fig. 3. Hardware architecture of the proposed classical VPU.

Stage 7 - Multi-Feature Fusion (ESC Fusion): Three parallel 3×3 kernels extract complementary features: a weighted Sobel variant ($[3 \ 10 \ 3]$ taps, $|G_x| + |G_y|$) for edges (E), local contrast relative to neighborhood mean for saliency (S), and unsharp masking with configurable gain for contrast enhancement (C). The feature maps are fused as:

$$\text{out} = \frac{w_e \cdot E + w_c \cdot C + w_s \cdot S}{256} + \frac{k \cdot (2E - 255)}{256} \quad (3)$$

where $w_e + w_c + w_s = 256$ and k controls an edge detail-boost term. All the products are 16-bit and are right-shifted by 8 to yield an 8-bit output.

Stage ordering follows signal-domain considerations: denoising precedes temporal smoothing so the EMA operates on a clean signal; temporal smoothing precedes histogram analysis for stable percentile estimates; gamma is applied last, after all linear-domain operations.

III. RESULTS

A. Hardware Implementation Results

The complete pipeline was synthesized, placed, and routed in TSMC 65 nm CMOS at 12.5 MHz with a 1.0 V supply, occupying 0.81 mm^2 within a $900 \mu\text{m} \times 900 \mu\text{m}$ footprint (Fig. 4). Total on-chip memory is 8.56 kB, dominated by the 4.8 kB temporal EMA frame buffer; spatial filtering stages use 160 B–1920 B line buffers. Total power is 9.5 mW.

B. Pipeline Evaluation

1) *Dataset:* We evaluate on subsets from two public datasets covering complementary scene types. **SANPO** [13] is a large-scale outdoor pedestrian dataset; we sample 202 frames spanning varied lighting and motion conditions, representative of street-navigation scenarios. **SUNRGBD** [14] is an indoor RGB-D collection; we draw 250 frames covering cluttered office and living-room environments. All frames are resized to 640×480 prior to processing.

2) *Baselines:* We compare against two spatial baselines that span standard prosthesis preprocessing.

Lanczos2 [15]: Lanczos-windowed-sinc downsampling to 80×60 without enhancement. **CLAHE** [16]: contrast-limited adaptive histogram equalization (4×4 tiles, clip limit 2.0). **Lanczos2+EMA** and **CLAHE+EMA** add the proposed pipeline’s Temporal EMA module (identical α schedule) to isolate spatial vs. temporal contributions.

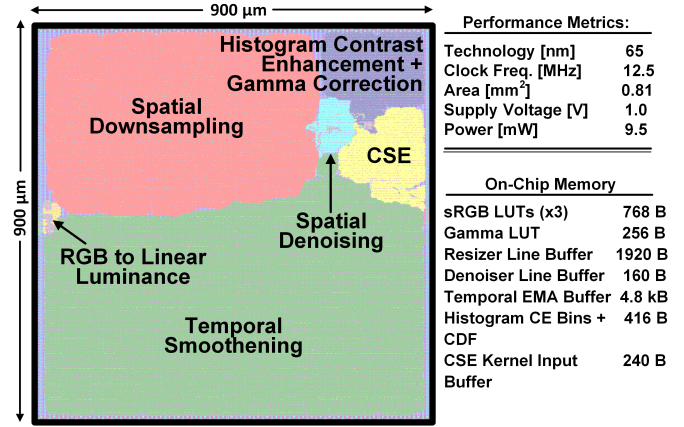


Fig. 4. Layout of the design implemented in 65nm CMOS and performance metrics with breakdown of the memory

3) *Phosphene Simulation:* All metrics are computed in the perceptual domain produced by the AxonMapModel [17] from the pulse2percept library [18], which uses the Janssonius nerve-fibre-bundle model to predict the elongated, streak-like phosphenes characteristic of epiretinal devices [19]. Array parameters match our hardware (80×60 disk electrodes, $75 \mu\text{m}$ pitch, $25 \mu\text{m}$ radius); model parameters are $\rho = 75 \mu\text{m}$ and $\lambda_{\text{ax}} = 200 \mu\text{m}$, with the visual-field grid sampled at 0.1° over $24^\circ \times 18^\circ$. Pixel values map linearly to electrode currents in $[0, 300] \mu\text{A}$.

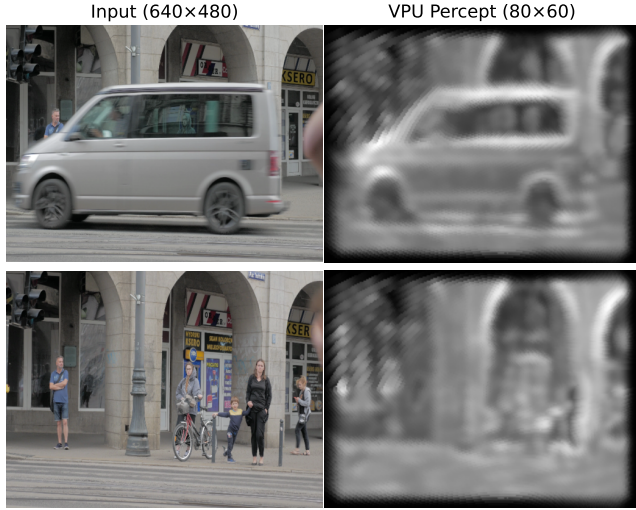


Fig. 5. Simulated phosphene percepts via the pulse2percept AxonMapModel. Left: raw input frame (640×480). Right: percept from VPU-processed output (80×60). Top: stationary vehicle; bottom: pedestrian scene.

4) *Metrics*: We report two complementary metrics, both computed on the post-AxonMap percept rather than on the 80×60 stimulus.

Saliency Contrast Ratio (SCR) is the per-frame metric. Let E be the binary mask of scene-edge pixels—the top 15% Sobel-magnitude pixels of the 640×480 input, dilated by 2 px and nearest-neighbour resampled to the percept grid. SCR is then

$$\text{SCR} = \frac{\overline{P[E]}}{\overline{P[\bar{E}]}} \quad (4)$$

where P is the percept and overbars denote spatial means. $\text{SCR} > 1$ indicates phosphene brightness is concentrated on scene-edge regions, directly measuring the goal of an edge-aware preprocessor.

Percept-domain flicker (CV) measures video stability. We add zero-mean Gaussian sensor noise ($\sigma=8$ LSB, $\sim 3\%$ full scale) to a static input for 30 time steps, render each method’s output through the AxonMapModel, and discard the first 10 frames for EMA steady-state. The remaining 20 frames yield a per-pixel temporal standard deviation σ_t , spatially averaged. We report $\text{CV}=\sigma_t/\mu$ where μ is mean percept brightness; by Weber’s law perceived flicker scales with brightness, so CV removes the mean-activation confound.

C. Results

Fig. 5 shows two representative scenes processed through the VPU pipeline, with the simulated phosphene percept alongside each raw input. Quantitative results are reported in Tables I and II. Win rate denotes the fraction of frames where the proposed pipeline exceeds Lanczos2; p -values are from paired one-sided Wilcoxon signed-rank tests against Lanczos2.

Per-frame quality: The proposed pipeline is the only method that exceeds $\text{SCR}=1$ on both datasets, reaching 1.054 indoors and 1.448 outdoors — a +24.8% improvement over Lanczos2 on SANPO and +21.0% on SUNRGBD. Both

TABLE I
PER-FRAME SCR IN THE PERCEPT DOMAIN. MEAN $\pm$ STANDARD DEVIATION ACROSS ALL FRAMES; COMBINED IS UNWEIGHTED MEAN ACROSS BOTH DATASETS

Method	SANPO	SUNRGBD	Combined
Lanczos2	1.160 $\pm$ 0.276	0.871 $\pm$ 0.173	1.016
CLAHE	1.145 $\pm$ 0.218	0.876 $\pm$ 0.160	1.011
Proposed	1.448 $\pm$ 0.401	1.054 $\pm$ 0.165	1.251
Win rate (%)	98.0	97.2	97.6
p -value (Wilcoxon)	7×10^{-35}	2×10^{-41}	—

TABLE II
TEMPORAL FLICKER IN THE PERCEPT DOMAIN ($\text{CV}=100 \cdot \sigma_t/\mu$).

Method	$\sigma_t \downarrow$	μ	CV (%) $\downarrow$
Lanczos2	1.13	108.2	1.05
CLAHE	4.86	126.9	3.83
Lanczos2+EMA	0.39	107.6	0.37
CLAHE+EMA	2.51	126.9	1.98
Proposed	0.26	97.2	0.27

baselines fall below 1 on SUNRGBD, indicating more current delivered to flat backgrounds than to scene structure; the proposed pipeline’s percentile-based contrast stretch and centre-surround edge weighting reverse this allocation. The method strictly outperforms Lanczos2 on 97.6% of all 452 frames (paired Wilcoxon $p < 10^{-34}$ per dataset).

Temporal stability: The proposed pipeline reduces percept-domain flicker by 1.4× over Lanczos2+EMA and 7.4× over CLAHE+EMA. Since these baselines share the proposed pipeline’s Temporal EMA module with identical parameters, the residual gap reflects spatial design: the separable 1:2:1 denoiser suppresses high-frequency noise before histogram analysis, and the percentile-tracking GlobalHistCE smooths its CDF parameters across frames so the contrast-stretch mapping does not bounce frame-to-frame. CLAHE’s tile-based stretch instead amplifies residual noise, yielding $\text{CV}=1.98\%$ even with EMA enabled.

IV. CONCLUSION

This paper presented a streaming, fixed-point seven-stage VPU pipeline for epiretinal prostheses, implemented in TSMC 65 nm CMOS, occupying 0.81 mm² and consuming 9.5 mW at 12.5 MHz with 8.56 kB of on-chip memory. Phosphene-domain evaluation showed the pipeline concentrates percept brightness on scene edges and achieves the lowest inter-frame flicker among compared methods, validating its suitability for prosthetic vision preprocessing. Future work will explore augmenting the classical pipeline with lightweight deep learning inference for improved scene adaptability.

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